Project: Worked examples
Title: Glaser condensation (interstitial and surface)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Interstitial condensation in a layered wall by the Glaser method over twelve monthly climate steps: the temperature and saturation pressure at every layer interface, the vapour-pressure line with its tangent construction at the condensation planes, the condensate on each interface month by month carried forward from the first month of condensation (drying when the weather turns), the annual balance, the peak accumulation against the owner's limit, and the surface temperature factor f_Rsi against its minimum at the critical surface humidity.

WHAT IT CALCULATES	RELATION
Layer resistance and diffusion thickness	$R_j = d_j/\lambda_j$; $s_{d,j} = \mu_j \cdot d_j$
Total resistance and interface temperature	$R_T = R_{si} + \Sigma\, R_j + R_{se}$; $\theta_k = \theta_i - (\theta_i - \theta_e)\cdot(R_{si} + \Sigma_{(j\le k)}\, R_j)/R_T$
Saturation pressure (Magnus)	$p_{sat} = 610.5\cdot\exp(17.269\cdot\theta/(237.3 + \theta))$ over water ($\theta \ge 0$ °C); $p_{sat} = 610.5\cdot\exp(21.875\cdot\theta/(265.5 + \theta))$ over ice ($\theta < 0$ °C)
Outside vapour pressure	$p_e = RH_e \cdot p_{sat}(\theta_e)$
Inside vapour pressure	$p_i = RH_i \cdot p_{sat}(\theta_i)$, or $p_e + f_{\Delta p}\cdot\Delta v(\theta_e)\cdot R_v\cdot(T_i + T_e)/2$ with the moisture excess Δv linear between two user points and constant beyond them (R_v = 461.5 J/(kg·K)) and the safety factor $f_{\Delta p}$ on the pressure difference (a record, default 1.10; clause wording to be checked against the owner's copy of ISO 13788)
Vapour flux through a layer	$g = \delta 0\cdot\Delta p/\Delta s_d$ with $\delta 0 = 2\times10^{-10}$ kg/(m·s·Pa)
Tangent construction	the line from p_i to p_e is bent onto the saturation line at the interfaces it would rise above (and at wet interfaces), by the smallest-slope rule from each contact point
Condensation at an interface	$g_c = g_{in} - g_{out}$; Mc accumulates $g_c\cdot t$ over 12 months from the first month of condensation (t = the row's days, else the calendar days of the month, January to December order), a wet interface (held at saturation) drying at $g_c < 0$ until nothing is left
Refinement check	the same year is run again with every layer split into 20 equal sub-layers of the same λ and μ (inside the model, not counted against the 30-layer limit), because the saturation line is curved inside a layer and the interface-only model can miss the condensate that forms between two interfaces; the annual balance passes only when it passes both ways, and the sheet warns when the two runs differ (a...

WHAT IT CALCULATES	RELATION
Annual balance and peak	PASS when no condensate remains after the 12 months (as entered and in 20 sub-layers per layer); Mc_{max} = the most held on any interface, checked against the user's limit
Wall U-value and its cross-check	$U = 1/R_T$; when the heat-loss calculation's U-value U_{ref} is given, the sheet warns if $\lvert U - U_{ref}\rvert/U_{ref}$ exceeds 2 % (the two calculations then describe different constructions)
Surface factor	$f_{Rsi} = 1 - R_{si,f}/R_{T,f}$ with $R_{si,f}$ = 0.25 m²K/W by default for an opaque element (conventional, a record; 0.13 for glazing); $f_{Rsi,min}$ = the largest monthly $(\theta_{si,min} - \theta_e)/(\theta_i - \theta_e)$ with $p_{sat}(\theta_{si,min}) = p_i/\varphi_{crit}$; PASS when $f_{Rsi} \geq f_{Rsi,min}$

Basis: Hygrothermal performance of building components and elements — internal surface temperature and surface condensation[1](cl. 5 (clause number to be checked against the owner's copy)); Hygrothermal performance of building components and elements — interstitial condensation (Glaser method)[1](cl. 6 (clause number to be checked against the owner's copy)); Hygrothermal performance of building components and elements — internal humidity from the humidity class[1](moisture excess and its design margin on Δp (clause number and wording to be checked against the owner's copy)); Hygrothermal performance of building components — assessment of moisture transfer by numerical simulation[2](whole standard (named as the recommended method for timber and breathable builds)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
layers	Wall layers from the inside outwards (d, λ, μ; the shape of the B4 layers table plus μ)	3 row(s)		drawing: A-310 rev B	any
Rsi	Internal surface resistance R_{si}	0.13	m²·K/W	spec: Project surface-resistance record (internal, horizontal flow), assumed	0.04 … 0.25 m²·K/W
Rse	External surface resistance R_{se}	0.04	m²·K/W	spec: Project surface-resistance record (external), assumed	0.02 … 0.25 m²·K/W
Rsi_f	Internal surface resistance $R_{si,f}$ for the f_{Rsi} check (0.25 m²K/W is the conventional value for an opaque element, a record to check against the owner's copy of ISO 13788; 0.13 for glazing)	0.25	m²·K/W	spec: Conventional R_si,f = 0.25 m²K/W for the surface-condensation check of an opaque element (a record, to be checked against the owner's copy of ISO 13788), assumed	0.04 … 0.5 m²·K/W
θ_{int}	Internal air temperature	20	°C	spec: SPEC-A-02 §3.1: living-room design temperature	10 … 30 °C

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$humidity_{mode}$	Internal humidity from	constant internal relative humidity		assumption: Constant internal humidity; no humidity-class record is held for this project.	any
RH_{int}	Internal relative humidity (constant-humidity mode)	50	%	spec: SPEC-A-02 §3.1: internal design humidity	20 ... 90 %
dp_f	Safety factor on the vapour-pressure difference $\Delta p = \Delta v \cdot R_v \cdot (T_i + T_e)/2$ in the moisture-excess mode (design margin, 1.10 by convention; a record, wording to be checked against the owner's copy of ISO 13788)	1.1		default	1 ... 1.5
climate	Twelve monthly outside conditions (the user's climate record, in the order the year is run)	12 row(s)		assumption: Invented illustrative maritime year; replace with the project's own monthly climate record (take it from your licensed source).	any
RH_{crit}	Critical surface relative humidity	80	%	spec: SPEC-A-02 §4.4: critical surface humidity	50 ... 100 %
Mc_{limit}	Limit on the condensate on any interface (blank = report only)	500	g/m²	spec: SPEC-A-02 §4.4: limit on the condensate per interface	1 ... 5000 g/m²
breathable	The build-up is timber or breathable (hygroscopic layers, capillary transport)	no		assumption: Mineral wool and brick: no timber or hygroscopic layers.	any

TABLE ROW	VALUES
layers[1]	name = Plasterboard; d = 12.5 mm; lambda = 0.25 W/(m*K); mu = 8
layers[2]	name = Mineral wool; d = 50 mm; lambda = 0.035 W/(m*K); mu = 1
layers[3]	name = Existing brick; d = 215 mm; lambda = 0.8 W/(m*K); mu = 10
climate[1]	month = M1; theta_e = 2 degC; RH_e = 90 %
climate[2]	month = M2; theta_e = 3 degC; RH_e = 88 %
climate[3]	month = M3; theta_e = 6 degC; RH_e = 84 %
climate[4]	month = M4; theta_e = 9 degC; RH_e = 80 %

TABLE ROW	VALUES
climate[5]	month = M5; theta_e = 13 degC; RH_e = 76 %
climate[6]	month = M6; theta_e = 16 degC; RH_e = 74 %
climate[7]	month = M7; theta_e = 18 degC; RH_e = 74 %
climate[8]	month = M8; theta_e = 18 degC; RH_e = 76 %
climate[9]	month = M9; theta_e = 15 degC; RH_e = 82 %
climate[10]	month = M10; theta_e = 11 degC; RH_e = 87 %
climate[11]	month = M11; theta_e = 6 degC; RH_e = 90 %
climate[12]	month = M12; theta_e = 3 degC; RH_e = 91 %

3. Assumptions

A1. Steady state in each month: one-dimensional diffusion of water vapour through homogeneous layers in series, with the monthly mean climate; no air movement, no liquid transport, no moisture storage, no solar or rain gain. (The Glaser method.)
A2. Material properties (λ, μ) are constants of the layer, not functions of its moisture or temperature; condensate does not change them. (The Glaser method.)
A3. Condensation is checked at the interfaces between layers; the saturation line is joined straight between the interface values, which can miss condensate that forms inside a layer. The annual balance is therefore also run with every layer split into 20 sub-layers and passes only when both pass; the limit on the condensate per interface is checked on the interfaces as entered. (Stated limit; refinement check (bs.glaser-01). Whether ISO 13788 requires sub-division is to be checked against the owner's copy.)
A4. The Magnus saturation pressure is used over water at and above 0 °C and over ice below, and the water-vapour permeability of air is $\delta_0 = 2\times10^{-10}$ kg/(m·s·Pa); both are closed-form physics, not data of a standard. (Derived.)
A5. Where a month's length is blank it is the calendar days of the month, taking the climate rows as January to December (February 28 days); enter the days where the record starts in another month. A wet interface that dries within a month is taken as dry from the next month; the month's rates are those of the wet state at its start. (Monthly steps.)
A6. The monthly climate, the humidity-class points, the surface resistances, the critical humidity and the limit on the condensate are the user's records from the user's own licensed copies; no climate data, humidity-class table, material table or limit of any standard is shipped. (No table from a standard is shipped.)
A7. $\text{humidity}_{\text{mode}}$ (Internal humidity from) is taken as constant internal relative humidity. (Constant internal humidity; no humidity-class record is held for this project.)
A8. climate (Twelve monthly outside conditions (the user's climate record, in the order the year is run)) is taken as 12 row(s). (Invented illustrative maritime year; replace with the project's own monthly climate record (take it from your licensed source).)
A9. breathable (The build-up is timber or breathable (hygroscopic layers, capillary transport)) is taken as no. (Mineral wool and brick: no timber or hygroscopic layers.)
A10. Month length: 12 row(s) of the climate gave no days, so the calendar days of the month are used with the rows taken as January to December (February 28 days). If the record starts in another month, enter the days of each row. (Calendar months.)

4. Working

[8] $R_1 = d / \lambda$ [1](cl. 6)
= Plasterboard: 12.5 mm / 0.25 W/(m·K)
= 0.05 m²·K/W
Plasterboard: thermal resistance

[9] $sd_1 = \mu \cdot d$ [1](cl. 6)
= Plasterboard: 8 × 12.5 mm
= 0.1 m
Plasterboard: diffusion-equivalent air layer thickness
[10] $R_2 = d / \lambda$ [1](cl. 6)
= Mineral wool: 50 mm / 0.035 W/(m·K)
= 1.4286 m²·K/W
Mineral wool: thermal resistance
[11] $sd_2 = \mu \cdot d$ [1](cl. 6)
= Mineral wool: 1 × 50 mm
= 0.05 m
Mineral wool: diffusion-equivalent air layer thickness
[12] $R_3 = d / \lambda$ [1](cl. 6)
= Existing brick: 215 mm / 0.8 W/(m·K)
= 0.26875 m²·K/W
Existing brick: thermal resistance
[13] $sd_3 = \mu \cdot d$ [1](cl. 6)
= Existing brick: 10 × 215 mm
= 2.15 m
Existing brick: diffusion-equivalent air layer thickness
[14] $R_T = R_{si} + \Sigma R_j + R_{se}$ [1](cl. 6)
= 0.13 + 1.7473 + 0.04
= 1.9173 m²·K/W
Total thermal resistance
[15] $U_{wall} = 1/R_T$ [1](cl. 6)
= 1 / 1.9173
= 0.52156 W/(m²·K)
U-value of the wall from the layers and the surface resistances above
[16] $S_d = \Sigma s_{d,j}$ [1](cl. 6)
= 2.3 m
= 2.3 m
Total diffusion-equivalent air layer thickness
[17] $psat_i = 610.5 \cdot \exp(17.269 \cdot \theta / (237.3 + \theta))$ [1](cl. 6)
= θ = 20 °C
= 2337 Pa
Saturation vapour pressure at the internal air temperature (Magnus)
[18] $p_{ext,m} = RH_e \cdot p_{sat}(\theta_e)$ [1](cl. 6)
= months 1 … 12 of the climate
= [634.76, 666.5, 785.09, 917.96, 1137.7, 1344.8, 1526.5, 1567.8, 1397.6, 1141.4, 841.16, 689
Outside vapour pressure by month
[19] $p_{int,m} = RH_i \cdot p_{sat}(\theta_i)$ [1](cl. 6)
= 50 % × 2337 Pa
= [1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 11
Inside vapour pressure by month
[20] n_{start} = first month of condensation [1](cl. 6)
= M11 (row 11)
= 11
Row of the climate in which the accumulation starts (0 when nothing condenses)
[21] $Mc_1 = \Sigma$ Mc over the interfaces [1](cl. 6)
= M1: θ_e = 2 °C; 1023.7 g/m² gained (-: dried) this month
= 2069.8 g/m²
M1: condensate on the wall at the end of the month

[22] Mc_2 = Σ Mc over the interfaces [1](cl. 6)
= M2: θ_e = 3 °C; 752.09 g/m² gained (-: dried) this month
= 2821.9 g/m²
M2: condensate on the wall at the end of the month
[23] Mc_3 = Σ Mc over the interfaces [1](cl. 6)
= M3: θ_e = 6 °C; 200.73 g/m² gained (-: dried) this month
= 3022.6 g/m²
M3: condensate on the wall at the end of the month
[24] Mc_4 = Σ Mc over the interfaces [1](cl. 6)
= M4: θ_e = 9 °C; -517.84 g/m² gained (-: dried) this month
= 2504.8 g/m²
M4: condensate on the wall at the end of the month
[25] Mc_5 = Σ Mc over the interfaces [1](cl. 6)
= M5: θ_e = 13 °C; -1698.2 g/m² gained (-: dried) this month
= 806.63 g/m²
M5: condensate on the wall at the end of the month
[26] Mc_6 = Σ Mc over the interfaces [1](cl. 6)
= M6: θ_e = 16 °C; -806.63 g/m² gained (-: dried) this month
= 0 g/m²
M6: condensate on the wall at the end of the month
[27] Mc_7 = Σ Mc over the interfaces [1](cl. 6)
= M7: θ_e = 18 °C; 0 g/m² gained (-: dried) this month
= 0 g/m²
M7: condensate on the wall at the end of the month
[28] Mc_8 = Σ Mc over the interfaces [1](cl. 6)
= M8: θ_e = 18 °C; 0 g/m² gained (-: dried) this month
= 0 g/m²
M8: condensate on the wall at the end of the month
[29] Mc_9 = Σ Mc over the interfaces [1](cl. 6)
= M9: θ_e = 15 °C; 0 g/m² gained (-: dried) this month
= 0 g/m²
M9: condensate on the wall at the end of the month
[30] Mc_{10} = Σ Mc over the interfaces [1](cl. 6)
= M10: θ_e = 11 °C; 0 g/m² gained (-: dried) this month
= 0 g/m²
M10: condensate on the wall at the end of the month
[31] Mc_{11} = Σ Mc over the interfaces [1](cl. 6)
= M11: θ_e = 6 °C; 207.78 g/m² gained (-: dried) this month
= 207.78 g/m²
M11: condensate on the wall at the end of the month
[32] Mc_{12} = Σ Mc over the interfaces [1](cl. 6)
= M12: θ_e = 3 °C; 838.33 g/m² gained (-: dried) this month
= 1046.1 g/m²
M12: condensate on the wall at the end of the month
[33] $Mc_{if,1}$ = max over the year of Mc [1](cl. 6)
= between 'Plasterboard' and 'Mineral wool'
= 0 g/m²
Most condensate held on the interface between 'Plasterboard' and 'Mineral wool'
[34] $Mc_{if,2}$ = max over the year of Mc [1](cl. 6)
= between 'Mineral wool' and 'Existing brick'
= 3022.6 g/m²
Most condensate held on the interface between 'Mineral wool' and 'Existing brick'

[35] Mc_{max} = max over interfaces and months of Mc [1](cl. 6)
= 3022.6 g/m²
= 3022.6 g/m²
Most condensate held on any one interface in the year
[36] Mc_{tot} = max over months of Σ Mc [1](cl. 6)
= 3022.6 g/m²
= 3022.6 g/m²
Most condensate on the whole wall in the year
[37] Mc_{end} = Σ Mc after 12 months [1](cl. 6)
= 0 g/m²
= 0 g/m²
Condensate left on the wall at the end of the 12-month cycle
[38] n_{diag} = month drawn
= M3 (row 3)
= 3
Row of the climate the diagram vectors are for
[39] sd_{if} = Σ s_d to the node [1](cl. 6)
= 4 nodes, inside surface first
= [0, 0.1, 0.15, 2.3] m
Diffusion thickness at the surfaces and interfaces
[40] x_{if} = Σ d to the node [1](cl. 6)
= 4 nodes
= [0, 0.0125, 0.0625, 0.2775] m
Depth from the inside surface at the surfaces and interfaces
[41] $\theta_{if} = \theta_i - (\theta_i - \theta_e)\cdot(R_{si} + \Sigma R)/R_T$ [1](cl. 6)
= θ_e = 6 °C
= [19.051, 18.686, 8.2544, 6.2921] degC
Temperature at the surfaces and interfaces
[42] $psat_{if}$ = 610.5·exp(17.269·θ/(237.3 + θ)), ice below 0 °C [1](cl. 6)
= 4 nodes
= [2203.1, 2153.5, 1090.9, 953.69] Pa
Saturation vapour pressure at the surfaces and interfaces (the Glaser saturation line)
[43] p_{if} = tangent construction [1](cl. 6)
= 4 nodes
= [1168.5, 1116.8, 1090.9, 785.09] Pa
Vapour pressure at the surfaces and interfaces after the tangent construction
[44] $R_{Tf} = R_{si,f} + \Sigma R_j + R_{se}$ [1](cl. 5)
= 0.25 + 1.7473 + 0.04
= 2.0373 m²·K/W
Total thermal resistance with the internal surface resistance of the surface check
[45] $fRsi = 1 - R_{si}/R_T$ [1](cl. 5)
= 1 - 0.25 / 2.0373
= 0.87729
Temperature factor of the internal surface
[46] $fRsi_{min}$ = max over months of $(\theta_{si,min} - \theta_e)/(\theta_i - \theta_e)$ [1](cl. 5)
= $\theta_{si,min}$ from p_i/φ_{crit}; governing month: M1
= 0.59026
Minimum temperature factor at the critical surface humidity
[47] n_{fcrit} = month that governs $f_{Rsi,min}$ [1](cl. 5)
= M1 (row 1)
= 1
Row of the climate that governs the surface check (0 when none applies)

[48] $Mc_{end,fine}$ = Σ Mc after 12 months, each layer in 20 sub-layers [1](cl. 6)
= 0 g/m²
= 0 g/m²
Condensate left after 12 months when every layer is split into 20 sub-layers (refinement check)
[49] $Mc_{tot,fine}$ = max over months of Σ Mc, each layer in 20 sub-layers [1](cl. 6)
= 3022.6 g/m²
= 3022.6 g/m²
Most condensate on the whole wall when every layer is split into 20 sub-layers (refinement check)
[51] u_{Mc} = Mc_{max} / Mc_{limit}
= 3022.6 g/m² / 500 g/m²
= 6.0452
Utilisation of the limit on the condensate per interface
[52] u_f = $f_{Rsi,min}$ / f_{Rsi} [1](cl. 5)
= 0.59026 / 0.87729
= 0.67282
Utilisation of the surface temperature factor
[53] pass = every check passes
= 3 check(s)
= 0
1 when the annual balance, the limit and the surface factor all pass, else 0

5. Results

SYMBOL	VALUE	DESCRIPTION
Mc_{max}	3022.6 g/m²	Most condensate on any interface
fRsi	0.87729	Temperature factor of the internal surface
pass	0	All checks pass
U_{wall}	0.52 W/(m²·K)	U-value of the wall
R_T	1.917 m²·K/W	Total thermal resistance
S_d	2.30 m	Total diffusion-equivalent air layer thickness
$psat_i$	2337 Pa	Saturation vapour pressure at the internal temperature
$p_{ext,m}$	[634.76, 666.5, 785.09, 917.96, 1137.7, 1344.8, 1526.5, 1567.8, 1397.6, 1141.4, 841.16, 689.22] Pa	Outside vapour pressure by month
$p_{int,m}$	[1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5] Pa	Inside vapour pressure by month
n_{start}	11	First month of condensation
Mc_1	2069.8 g/m²	M1: condensate on the wall
Mc_2	2821.9 g/m²	M2: condensate on the wall
Mc_3	3022.6 g/m²	M3: condensate on the wall
Mc_4	2504.8 g/m²	M4: condensate on the wall
Mc_5	806.6 g/m²	M5: condensate on the wall
Mc_6	0 g/m²	M6: condensate on the wall

SYMBOL	VALUE	DESCRIPTION
Mc_7	0 g/m²	M7: condensate on the wall
Mc_8	0 g/m²	M8: condensate on the wall
Mc_9	0 g/m²	M9: condensate on the wall
Mc_{10}	0 g/m²	M10: condensate on the wall
Mc_{11}	207.8 g/m²	M11: condensate on the wall
Mc_{12}	1046.1 g/m²	M12: condensate on the wall
$Mc_{if,1}$	0 g/m²	Peak condensate on interface 1
$Mc_{if,2}$	3022.6 g/m²	Peak condensate on interface 2
Mc_{tot}	3022.6 g/m²	Most condensate on the wall
Mc_{end}	0 g/m²	Condensate left after 12 months
$Mc_{end,fine}$	0 g/m²	Condensate left after 12 months, each layer in 20 sub-layers
$Mc_{tot,fine}$	3022.6 g/m²	Most condensate on the wall, each layer in 20 sub-layers
n_{diag}	3	Month of the diagram
sd_{if}	[0, 0.1, 0.15, 2.3] m	Diffusion thickness at the nodes
x_{if}	[0, 0.0125, 0.0625, 0.2775] m	Depth at the nodes
θ_{if}	[19.051, 18.686, 8.2544, 6.2921] degC	Temperature at the nodes
$psat_{if}$	[2203.1, 2153.5, 1090.9, 953.69] Pa	Saturation pressure at the nodes
p_{if}	[1168.5, 1116.8, 1090.9, 785.09] Pa	Vapour pressure at the nodes
$fRsi_{min}$	0.59026	Minimum temperature factor
n_{fcrit}	1	Month governing the surface check
R_{Tf}	2.037 m²·K/W	Total resistance for the surface check
u_{Mc}	6.05	Peak condensate against its limit (governing)
u_f	0.67	Surface factor against its minimum

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Peak condensate on an interface against its limit (500 g/m²)	u_{Mc} = 6.05	u_{Mc} ≤ 1	6.05	[1](cl. 6)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Surface condensation: f_{Rsi} against $f_{Rsi,min}$ at 80 % critical humidity	u_f = 0.67	u_f ≤ 1	0.67	[1](cl. 5)
PASS	Annual moisture balance: the condensate dries out within the year (as entered and with every layer in 20 sub-layers)	Mc_{end} = 0 g/m² as entered, 0 g/m² in 20 sub-layers per layer	Mc_{end} ≤ 0 g/m²	–	[1](cl. 6)

7. Applicability

WARNING records: tabulated values not taken from a reference record are layers (Layer conductivity and vapour resistance); Rsi, Rse, Rsi_f (Surface resistances); climate (Monthly outside climate); RH_{crit}, Mc_{limit} (Condensation criteria). Enter each as one so its source, edition and licence travel with it. [record]

Defaults used (inputs left empty):
dp_f: not entered, so the default of 1.1 is used.

8. Verification

Template bs.glaser@1.3.0 Content hash bfa122129a1b6cf7 Engine calc-engine/1.0.0
Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. ISO 13788.
2. EN 15026.